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PHYSICS CONTRIBUTION

COMPARISON BETWEEN IN-BEAM AND OFFLINE POSITRON EMISSION TOMOGRAPHY IMAGING OF PROTON AND CARBON ION THERAPEUTIC IRRADIATION AT SYNCHROTRON- AND CYCLOTRON-BASED FACILITIES

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Purpose: The benefit of using dedicated in-beam positron emission tomography (PET) detectors in the treatment room instead of commercial tomographs nearby is an open question. This work quantitatively compares the measurable signal for in-beam and offline PET imaging, taking into account realistic acquisition strategies at different ion beam facilities. Both scenarios of pulsed and continuous irradiation from synchrotron and cyclotron accelerators are considered, because of their widespread use in most carbon ion and proton therapy centers.
Methods and Materials: A mathematical framework is introduced to compare the time-dependent amount and spatial distribution of decays from irradiation-induced isotope production. The latter is calculated with Monte Carlo techniques for real proton treatments of head-and-neck and paraspinal tumors. Extrapolation to carbon ion irradiation is based on results of previous phantom experiments. Biologic clearance is modeled taking into account available data from previous animal and clinical studies.
Results: Ratios between the amount of physical decays available for in-beam and offline detection range from 40% to 60% for cyclotron-based facilities, to 65% to 110% (carbon ions) and 94% to 166% (protons) at synchrotron-based facilities, and increase when including biologic clearance. Spatial distributions of decays during irradiation exhibit better correlation with the dose delivery and reduced influence of biologic processes.
Conclusions: In-beam imaging can be advantageous for synchrotron-based facilities, provided that efficient PET systems enabling detection of isotope decays during beam extraction are implemented. For very short (<2 min) irradiation times at cyclotron-based facilities, a few minutes of acquisition time after the end of irradiation are needed for counting statistics, thus affecting patient throughput. © 2008 Elsevier Inc.

Ion therapy, Positron emission tomography.

INTRODUCTION

Positron emission tomography (PET) has been suggested by several investigators as a valuable tool for *in vivo,* noninvasive confirmation of the treatment delivery in ion beam therapy (1–6). The method exploits the β^+-activity produced in nuclear interactions between the ions and the irradiated tissue. Phantom experiments (7–10) and clinical trials (11, 12) performed during (in-beam) or shortly after (offline) irradiation with carbon ion and proton beams indicate 1- to 2-mm achievable accuracy for verification of the beam range in rigidly immobilized sites. Use of PET imaging can also enable within-millimeter control of the lateral field position (1, 6, 8–10) and detection of deviations between planned and applied treatment, *e.g.,* because of minor misalignments or local changes of the patient anatomy or physiology (13). Therefore several institutions are considering the establishment of PET quality assurance at existent or upcoming ion beam therapy facilities worldwide. The optimal implementation is still a topic of debate.

Similar to the first prototype (14), in-beam PET detectors inside the treatment room are the method of choice for acquisition in treatment position—ideally in real-time during irradiation—and for detection of the activity contribution from short-lived emitters such as ^{15}O (half-life $T_{1/2}$=121.8s). This solution is clinically implemented at the Gesellschaft für Schwerioneneforschung (GSI), Darmstadt, Germany, for therapy with stable carbon ion beams (2, 11). A similar experimental installation was developed at the secondary beam course of the Heavy Ion Medical Accelerator in Chiba, Japan, for range verification of implanted radioactive beams

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(15). A third is under development in Kashiwa, Japan, for verification of proton treatments (9).

As a technical challenge, integration into the treatment environment poses geometric constraints because of the need of an opening for the beam portal (and exit cone of light fragments for heavy ions) and flexible patient positioning, typically resulting in dual-head configurations. Furthermore production of background radiation during beam extraction (2, 3, 16) and sensitivity of PET imaging to the time course of irradiation (especially for dynamic beam application) demand synchronization of the PET data acquisition with the beam control system (11). To date, the former problem restricts usable ''in-beam'' detection during the pauses of pulsed beam delivery (2, 3, 11) or directly after continuous irradiation (3, 9), although solutions have already been proposed and successfully tested (17).

A technically less demanding alternative is offered by commercial full-ring PET scanners, measuring the activity contribution from long-lived emitters such as ^{11}C ($T_{1/2}$=1223s) shortly after treatment (offline). This approach uses resources which are typically located outside the treatment room and can serve other applications besides ion beam quality assurance. The method becomes particularly attractive with devices combining PET and computed tomography (CT) (10, 12). The additional CT enables accurate co-registration between the treatment and imaging position, compensating for unavoidable patient movement caused by transportation and/or repositioning. Moreover it allows detection of morphologic modifications with respect to the planning CT, which is used in in-beam PET imaging for calculation of the attenuation map and co-registration to the patient anatomy.

The imaging performances are mostly conditioned by the geometrical detector arrangement and the severity of limited-angle artifacts (18). The second major factor is counting statistics (18), which, for a given fractionation scheme and detector arrangement, depends on the time course of the beam delivery and PET acquisition. In both in-beam and offline imaging scenarios, prolongation of the measuring time for the sake of statistics is limited. For detectors inside the treatment room, this results from the requirement of efficient patient throughput. For offline imaging at remote sites, major restrictions come from considerations of patient comfort. Finally biologic clearance and/or transport of the produced activity increases with the time after treatment, affecting the spatial distribution and the intensity of the measurable signal.

At the upcoming Heidelberg Ion-Beam Therapy (HIT) Centre in Heidelberg, Germany, we are evaluating the integration of the in-beam PET scanner, which has been developed and operated at the GSI experimental therapy unit by the Forschungszentrum Dresden-Rossendorf (11). In comparison to GSI, the dedicated synchrotron of the new hospital-based facility is designed for more efficient use of its duty cycle in combination with the active beam delivery system. This will result in shorter treatment times with reduced pauses during the pulsed beam delivery, compromising the counting statistics of in-beam PET imaging.

Therefore, this study aims to investigate the advantage of in-beam against offline PET imaging for each individual ion species (^{1}H, ^{12}C) and accelerator system. In addition to the synchrotron-based facilities at GSI and HIT, we include conventional cyclotrons coupled to passive beam shaping systems as widely used in most proton therapy centers. There is, however, no aim to compare, *e.g.,* PET imaging of synchrotron-based carbon ion treatments against protons at cyclotron-based facilities.

METHODS AND MATERIALS

Production of β^+-emitters

Amount and spatial distribution of irradiation-induced β^+-emitters were obtained from CT-based Monte Carlo (MC) calculations (19) for head-and-neck and paraspinal tumors treated with proton beams at Massachusetts General Hospital (MGH) in Boston, MA (12). This approach provides more accurate isotope identification, at the expense of computing time (19), in comparison to carbon ion modeling as described by Pönisch *et al.* (20), which adapts β^+-emitter yield in polymethylmethacrylate ($C_5H_8O_2$) to the patient anatomy by means of stretching and weighting algorithms. Moreover the β^+-activity pattern induced by light ions like protons is more sensitive to the influence of the time course of beam delivery and data acquisition. This results from the limitation to β^+-activation of target fragments of the irradiated tissue (8). Besides the production channels of ^{11}C, ^{15}O, ^{13}N, ^{38}K, ^{30}P, and ^{14}O considered by Parodi *et al.* (19), ^{10}C yield from proton interaction on ^{12}C was included based on the experimental cross-sections described elsewhere (21, 22).

The integral isotope production induced by carbon ion beams was extrapolated from the proton clinical cases, based on experimental β^+-emitter yield after proton and carbon ion irradiation of organic targets at the same penetration depth. According to data from Parodi *et al.* (5) and Fiedler *et al.* (23), the scaling formalism of Sihver *et al.* (24), and additional MC calculations, a ratio of 2.5 was assumed between the production of target fragments induced by the same amount of primary carbon ions and protons at the same range. An additional average-enhancing factor of 2 and 4 was applied to the yield of ^{11}C and ^{10}C induced by ^{12}C irradiation, to account for the possible formation as target and projectile fragments. The yield induced by the same amount of primary carbon ions was finally scaled down with the 20-fold lower fluence required for delivery of the same physical dose as protons (4). Normalization to the same biologic dose would require a further downscaling by the ratio $r \approx 1 - 3$ between the relative biologic effectiveness of carbon ions and protons. However this factor was omitted because it can vary substantially and is still subject to clinical debate. Note that this simplification does not affect our relative comparison between in-beam and offline PET imaging of carbon ion data. A detailed direct comparison between proton and carbon ion induced activation is beyond the scope of this work.

Mathematical framework

The number of isotopes decayed in different acquisition windows was modeled under the assumption of constant activation during irradiation of fixed parameters. This approximation disregards, *e.g.,* temporal variations of the beam energy, position, and intensity, as well as the adjustment of individual pulse lengths in the active beam delivery at the GSI and HIT synchrotrons. Similarly it neglects minor dynamic effects of passive modulation, *e.g.,* via a rotating wheel, for most cyclotron-based proton centers.

The advantage is to allow a considerable simplification of the mathematical formalism without impairing the comparative goal of this study.

Synchrotron. Similar to the report by Parodi *et al.* (5), a recursive model was used to describe the isotope build-up and decay during a synchrotron-based beam delivery, consisting in repetition of pulses (spills) of duration t_s followed by pauses of duration t_p at a fixed extraction period t_{extr}=t_s+t_p (Fig. 1a). The main ingredients are (a) the formation and decay of activity A_j because of the constant production rate K_j of β^+-emitters j of decay constant λ_j during irradiation, which is expressed in the first spill by:

$$A_j(t) = K_j\left(1 - \mathrm{e}^{-\lambda_j t}\right) \quad 0 \le t \le t_s, \tag{1}$$

(b) the subsequent activity decay between the beam pulses, *e.g.,* during the first pause:

$$A_j(t) = K_j\left(1 - \mathrm{e}^{-\lambda_j t_s}\right)\mathrm{e}^{-\lambda_j (t - t_s)} \quad t_s < t < t_{extr,} \tag{2}$$

and (c) the superimposition between newly formed activity and decays from activity produced in previous beam pulses for the spills after the first extraction cycle. The production rate K_j of each j species is given by the MC-calculated isotope production P_j divided by the total beam-on time, *i.e.,* $K_j = P_j / (\mathcal{S} \cdot t_s)$ for $\mathcal{S}$ spills.

The number N_j of decays from isotopes of species j in an acquisition interval Δt starting at a time t can be finally deduced (assuming a 100% β^+-decay branching ratio) as:

$$N_j(t, \Delta t) = \int_t^{t+\Delta t} A_j(t')dt' \tag{3}$$

Following this formalism, the number of in-beam decays from a given isotope species j can be split into the component $N_{j,spill}$ decayed during the $\mathcal{S}$ beam pulses:

$$N_{j,\mathrm{spill}} = \mathcal{S}D_{spill} + \frac{K_j}{\lambda_j}(1 - \mathrm{e}^{-\lambda_j t_s})^2 \mathrm{e}^{-\lambda_j t_p} \sum_{i=1}^{\mathcal{S}-1} (\mathcal{S} - i)\mathrm{e}^{-(i-1)\lambda_j t_{extr}}, \text{ where}$$
$$D_{spill} = K_j t_s - \frac{K_j}{\lambda_j}(1\text{-}\mathrm{e}^{-\lambda_j t_s}), \tag{4}$$

and into the component $N_{j,pause}$ decayed during the $\mathcal{S}$ -1 beam pauses:

$$N_{j,\mathrm{pause}} = \frac{K_j}{\lambda_j}\left(1 - \mathrm{e}^{-\lambda_j t_s}\right)\left(1 - \mathrm{e}^{-\lambda_j t_p}\right) \sum_{i=1}^{\mathcal{S}-1} (\mathcal{S} - i)\mathrm{e}^{-(i-1)\lambda_j t_{extr}} \tag{5}$$

If the acquisition is prolonged for a short time t_{end} directly after the end of irradiation (Fig. 1a), as it is done at GSI in the ~40 s needed by the medical staff to enter the treatment site for changing the patient position (11), additional decays $N_{j,end}$ can be collected:

$$N_{j,\mathrm{end}} = \frac{K_j}{\lambda_j}\left(1 - \mathrm{e}^{-\lambda_j t_s}\right)\left(1 - \mathrm{e}^{-\lambda_j t_{\mathrm{end}}}\right) \sum_{i=0}^{\mathcal{S}-1} \mathrm{e}^{-i\lambda_j t_{\mathrm{extr}}} \tag{6}$$

Similarly the amount $N_{j,off}$ of decays from isotopes of species j in an offline acquisition window t_{meas} starting with a delay t_{del} after end of irradiation is:

$$N_{j,\mathrm{off}} = \frac{K_j}{\lambda_j}\left(1 - \mathrm{e}^{-\lambda_j t_s}\right)\mathrm{e}^{-\lambda_j t_{\mathrm{del}}}\left(1 - \mathrm{e}^{-\lambda_j t_{\mathrm{meas}}}\right) \sum_{i=0}^{\mathcal{S}-1} \mathrm{e}^{-\mathrm{i}\lambda_j t_{\mathrm{extr}}} \tag{7}$$

Cyclotron. The formulas for PET imaging at a continuous cyclotron-based irradiation (Fig. 1b) can be derived from Eq. 4, 6, and 7 of a synchrotron-based beam delivery with only one spill of duration t_s equal to the total irradiation time t_{irr} (i.e., $\mathcal{S}$ =1 and t_{extr}=t_s=t_{irr}).

Biologic decay

Biologic processes of radiation-induced β^+-activity are a complex problem because of the unknown molecular forms. Nonetheless the formalism of Mizuno *et al.* (25), in combination with data from animal studies with stable or radioactive carbon ion beams (25, 26) and their empirically driven extension to human tissue, provided reasonable results in clinical offline PET/CT imaging after proton radiotherapy (12). Because of the 14 to 20 min elapsed between irradiation and imaging, that clinical study was sensitive only to the slow component of biologic washout (biologic half-life $T_{1/2bio,s}$~3,000–10,000 s). Thus the modeling described by Parodi *et al.* (12) is here extended to include the documented existence of fast ($T_{1/2bio,f}$~2–10 s) and medium ($T_{1/2bio,m}$~100–200 s) processes (25).

The MC-simulated isotope production P_j(r) is decomposed into three fractions M_l(r)P_j(r) ($\sum_l M_l$(r)=1 $\forall$r) undergoing fast (*f*), medium (*m*), and slow (*s*) biologic decay with a biologic half-life $T_{1/2,bio,l}$(r). Note that M_l(r) and $T_{1/2,bio,l}$(r) are regarded as tissue specific (*i.e.,* space dependent), and distinctions in the metabolism of individual isotope species j are neglected (25). Similar to the report by Parodi *et al.* (12), thresholds were set on the planning CT for identification of fat ($-150 \le$ Hounsfield Units [HU] ≤ -30), soft bone (200 $\le$ HU < 1,000), and compact bone (HU $\ge$1,000). In these tissues, the values of Parodi *et al.* (12) were used for the slow components of the biologic parameters, whereas assumptions based on trends in literature data (25, 26) were used for the fast and medium components. In all the remaining tissues, the animal data of brain

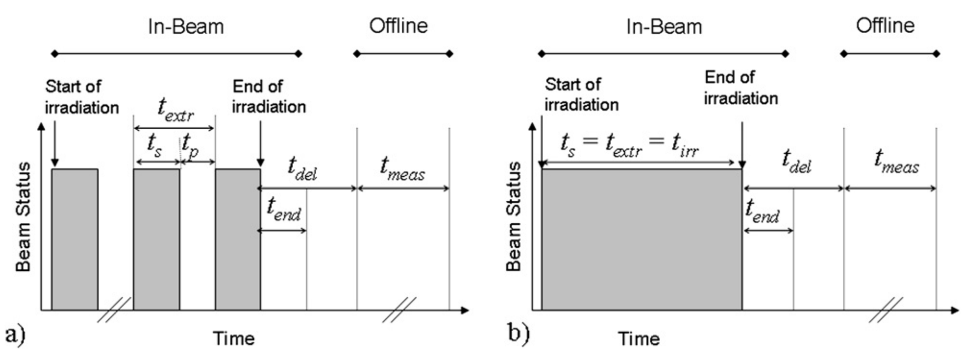

Fig. 1. Considered time structure of irradiation delivered by a synchrotron (left, a) and a continuous-wave cyclotron (right, b). Grey squares indicate beam-on status. Time intervals *t* correspond to definitions given in text. ''In-beam'' PET imaging includes data decayed during and shortly after irradiation. Offline acquisition is started with a delay t_{del} after irradiation.

Table 1. Values used for the fast (*f*), medium (*m*) and slow (*s*) biologic decay properties (*M*: fraction, $T_{1/2,\text{bio}}$: biologic half-life) of different tissue types

	Fast decay		Medium decay		Slow decay	
Tissue type	M_f	$T_{1/2,\ \text{bio},f}$ (s)	M_m	$T_{1/2,\ \text{bio},m}$ (s)	M_s	$T_{1/2,\ \text{bio},s}$ (s)
Hard bone	0.05	20	0.05	300	0.9[‡]	15,000[‡]
Soft bone	0.2	15	0.2	250	0.6[‡]	8,000[‡]
Fat	0.05	20	0.05	300	0.9[‡]	15,000[‡]
Muscle	0.3*	10*	0.15*	195*	0.55*	3,500*
Brain	0.35*	2*	0.3*	140*	0.35*	10,000*

Values marked by *were approximated from [25], whereas [‡]refers to [12].

and thigh muscle from Mizuno *et al.* (25) were used for head and paraspinal sites, respectively (Table 1).

To account for the competing processes of activity formation (during irradiation) and physical as well as biologic decay, the basic equations (1) and (2) have to be modified into:

$$A_j(\mathbf{r},t) = \sum_{l=1}^{3} \frac{\lambda_j}{\lambda_j + \lambda_{\text{bio},l}(\mathbf{r})} M_l(\mathbf{r}) K_j(\mathbf{r}) \left(1 - e^{-\left(\lambda_j + \lambda_{\text{bio},l}(r)\right)t}\right) \quad 0 \le t \le t_s, \tag{8}$$

$$A_j(\mathbf{r},t) = \sum_{l=1}^{3} \frac{\lambda_j}{\lambda_j + \lambda_{\text{bio},l}(\mathbf{r})} M_l(\mathbf{r}) K_j(\mathbf{r}) \left(1 - e^{-\left(\lambda_j + \lambda_{\text{bio},l}(\mathbf{r})\right)t_s}\right) e^{-(\lambda_j + \lambda_{\text{bio},l}(\mathbf{r}))t} \quad t_s < t < t_{\text{extr},} \tag{9}$$

where $\lambda_{\text{bio,l}}(\text{r}) = \log_e(2)/T_{1/2,\text{bio,l}}(\text{r})$. It follows that for each *l* phase of biologic decay, the formulae (4–7) still hold, provided that the substitutions $\lambda_j \rightarrow \lambda_j + \lambda_{\text{bio,l}}$ and $K_j \rightarrow M_l K_j \lambda_j / (\lambda_j + \lambda_{\text{bio,l}})$ are introduced. Thus, the total number of β^+-decays under consideration of biologic washout can be obtained from summation of the results individually calculated for the fast, medium, and slow processes, using the proposed modification of formulas (4–7) with the tissue-dependent parameters shown in Table 1.

RESULTS

Production of β^+-emitters

The MC-calculated dose distributions of individual portals are shown in Fig. 2 for clinical cases of skull-base and paraspinal tumors treated with multi-field irradiation at the proton therapy unit at MGH. The selected fields were chosen because they do not necessarily require a beam gantry, which, for example, is not available at GSI. The data were normalized to the delivery of a typical field dose of 1 Gy.

The corresponding spatial distributions of the most abundant positron emitters are depicted in Figs. 3 and 4 in the same isocenter coronal slice of Fig. 2. This choice corresponds to the plane which is imaged with best quality by dual-head PET scanners adapted to treatment sites with horizontal beam delivery and rotating table (11). The integral amount of all the considered β^+-emitters is summarized in Table 2, together with extrapolation to the carbon ion case.

Comparison of detectable decays for different beam delivery and imaging strategies

Irradiation and PET acquisition. Table 3 summarizes the time parameters that have been inserted in equations (4–7) for determination of the measurable decays from the calculated isotope production.

For in-beam PET imaging at GSI, the current acquisition strategy was taken into account, which uses events decayed during the synchrotron pauses and in the 40 s immediately after the end of the irradiation (11). From the clinical experience, the average treatment time per field was set to 10 min, consisting of 120 spills of 1-s duration followed by 4-s pauses.

For in-beam acquisitions at the upcoming HIT facility, 90% recovery of the events decayed during the beam pulses was assumed to be possible by implementation of the technical solution of Crespo *et al.* (17). Besides standard detection of decays in the beam pauses, the measuring time directly after treatment was increased to 60 s. This is a compromise between the need for high patient throughput and the reduction of statistics counting because of the shorter acquisition time. The latter is caused by the more efficient use of the accelerator and active beam delivery system, which was taken into account by assuming 3-min irradiation consisting in repetitions of 1.5-s spills and 1.5-s pauses.

The PET imaging of treatments delivered by continuous-wave cyclotrons is limited more acutely from the problem

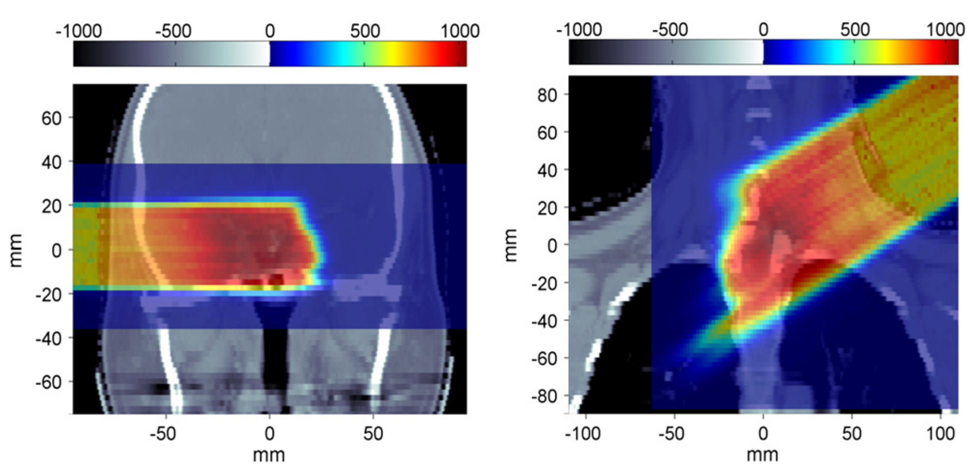

Fig. 2. Computed tomography (CT)–based Monte Carlo (MC) calculations of the planned proton dose for a lateral and an oblique portal delivered to a skull-base (left) and a T-spine (right) tumor. The MC output was normalized to a typical fraction dose of 1Gy (color bar in mGy). The planning CT is shown in grey scale, arbitrarily normalized for display purposes.

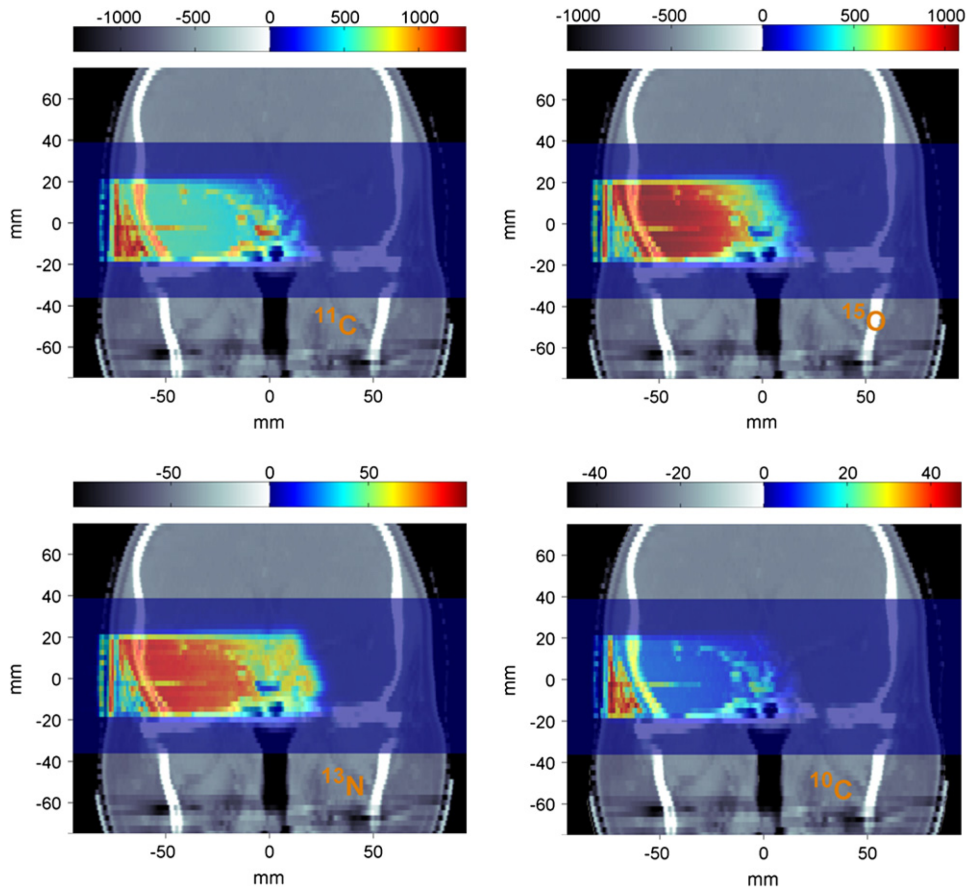

Fig. 3. Calculated spatial distribution of the most abundant ^{11}C, ^{15}O, ^{13}N, and ^{10}C positron emitters produced by 1-Gy proton irradiation of a skull-base tumor (Fig. 2, left). Color bar displays isotope concentrations in units of 10^3/ml. Grey scale displays arbitrarily rescaled CT numbers.

of background production during ion extraction, making ''in-beam'' acquisition impossible unless the solution of Crespo *et al.* (17) is implemented. The duration itself of the irradiation, which determines the available time window for ''in-beam'' detection, is also typically very short. Based on experience from MGH (10, 12), it was set to 60 s. To investigate the maximum achievable signal, the calculations were done for the ideal assumption of complete (*i.e.,* 100%) recovery of decays occurred during beam extraction. Prolongation of the acquisition directly after end of irradiation was limited to 40 s, however, which already corresponds to 67% of the beam delivery time.

For offline PET imaging, 30-min acquisition, starting 5 min after application of the treatment field, was assumed in all scenarios.

Quantitative yield of measurable decays. The spatially integrated amount of physical (*i.e.,* neglecting biologic considerations) decays from the main contributing isotopes is separately shown in Figs. 5 and 6 for the different irradiation and imaging modalities. The total amount from the entire isotope yield is summarized in Table 4.

These data indicate that for an irradiation time shorter than the 2-min half-life of ^{15}O as in the considered cyclotron case, even the maximum (*i.e.,* assuming ideal full recovery of the in-beam decays) measurable in-beam PET signal is significantly lower (40–60% in the considered example) than that detectable over several minutes in offline imaging shortly after treatment.

For the typically longer treatment times of synchrotron-based facilities with active beam delivery, the number of measurable decays during online imaging is comparable to or even larger than (up to a factor of $\sim$1.7) that detectable offline over a 3- to 10-fold longer acquisition window starting just 5 min after end of irradiation in the proton case. This is because about 70% to 85% of the decays collected in both in-beam and offline acquisition is caused by the main yield of ^{15}O and ^{11}C, respectively, which turned out to be very similar for the considered proton treatments (Figs. 5 and 6; Table 2). In the case of carbon ion irradiation, the increased ratio of $^{11}C/^{15}O$ production penalizes the available signal for online detection in comparison to offline PET imaging. This is especially evident for the shorter irradiation times achievable at the HIT facility. In the latter scenario, a ratio of 65% to 75% was found between the amount of detectable decays during in-beam and offline PET acquisitions, against the value of 97% to 110% obtained for delivery of the same carbon ion treatments at the GSI synchrotron.

Changes of the integral quantitative estimates resulting from biologic considerations are shown in Table 5 for proton irradiation. According to these calculations, biologic clearance is found to reduce the measurable signal by about

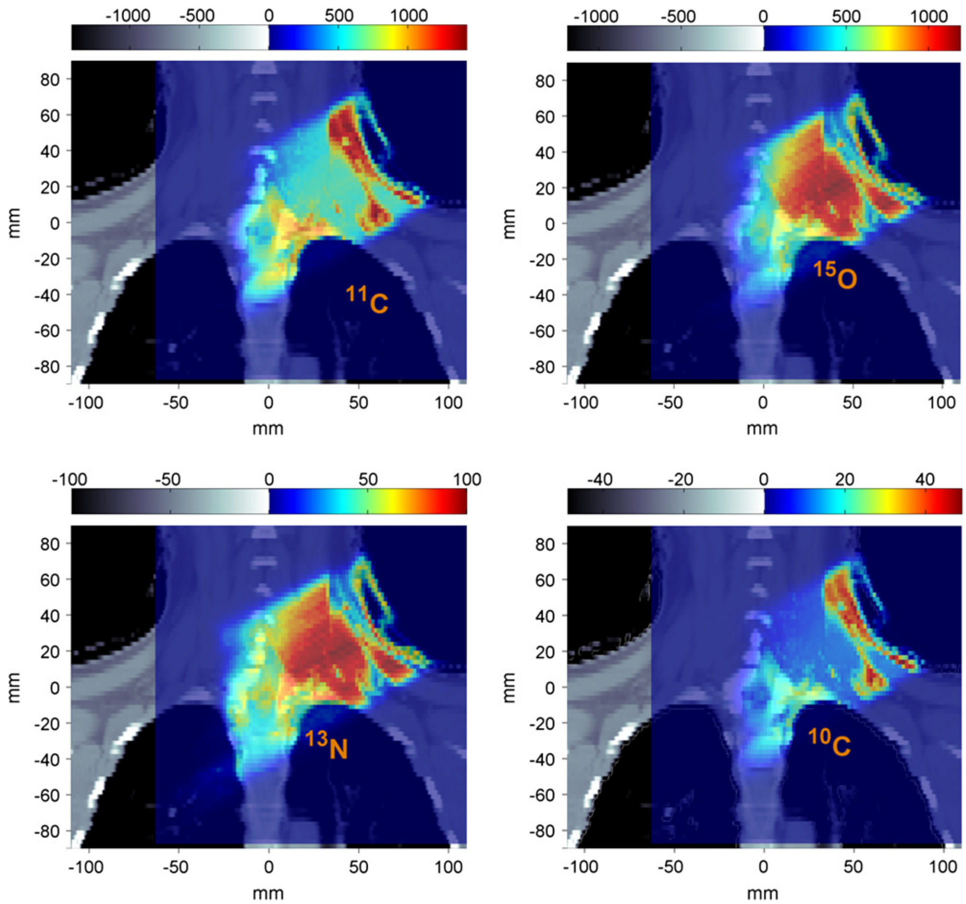

Fig. 4. Calculated spatial distribution of the most abundant ^{11}C, ^{15}O, ^{13}N, and ^{10}C positron emitters produced by 1-Gy proton single-field irradiation of a T-spine chondrosarcoma (Fig. 2, right). Interpretation of the grey/color bar is the same as in Fig. 3.

30% to 40% and by 20% to 30% for in-beam imaging of head and paraspinal sites, respectively. The corresponding reduction for offline imaging amounts to about 55% and 45%. This results in a less unfavorable ratio of about 70% to 90% (against 50–60%without biologic considerations) between the amount of decays which can be detectable in both in-beam and offline PET acquisitions at a cyclotron-based proton facility. For proton synchrotron-based facilities, a detectable signal up to a factor of 2.3 higher (vs. 1.7 without biologic considerations) is expected for in-beam PET imaging in comparison to offline acquisitions.

With respect to our main question on the applicability of in-beam PET at the more efficient HIT synchrotron, only a 20% (15% with biologic considerations) reduction of the detectable in-beam signal is observed in comparison to the GSI scenario, despite the decrease of the overall treatment time by a factor of 3.3 and the increase of the accelerator duty cycle from 20% to 50%. These estimations however demand the recovery of a large fraction (90%) of data decayed during the increased beam-on time via implementation of the random suppression method of Crespo *et al*. (17), which is currently unavailable at GSI. At the Heidelberg facility, the decays during the spills are expected to contribute equally to the measurable signal as those occurred during the accelerator pauses and in the 60 s immediately after the irradiation ends (Table 4). According to the ratio of spill and pause durations within each extraction cycle (Table 3), the signal detectable during the beam-on time accounts up to half of the entire measurable in-beam PET signal during irradiation, as opposed to the 20% potential contribution for the corresponding irradiation at GSI. Moreover an additional 20 s of acquisition after the end of irradiation can raise the measurable

Table 2. Calculated integral yield of β^+-emitters produced by proton and carbon ion delivery of the treatment fields shown in Fig. 2

Ion	Site	^{11}C	^{15}O	^{13}N	^{10}C	^{14}O	^{30}P	^{38}K
^{1}H	Head	$7.7\ 10^{7}$	$9.4\ 10^{7}$	$9.4\ 10^{6}$	$1.6\ 10^{6}$	$5.5\ 10^{5}$	$6.3\ 10^{5}$	$7.4\ 10^{5}$
^{1}H	Spine	$2.3\ 10^{8}$	$2.3\ 10^{8}$	$2.3\ 10^{7}$	$5.2\ 10^{6}$	$1.5\ 10^{6}$	$2.5\ 10^{6}$	$3.1\ 10^{6}$
^{12}C	Head	$1.9\ 10^{7}$	$1.2\ 10^{7}$	$1.2\ 10^{6}$	$8.1\ 10^{5}$	$6.9\ 10^{4}$	$7.9\ 10^{4}$	$9.3\ 10^{4}$
^{12}C	Spine	$5.9\ 10^{7}$	$2.8\ 10^{7}$	$2.9\ 10^{6}$	$2.6\ 10^{6}$	$1.9\ 10^{5}$	$3.1\ 10^{5}$	$3.9\ 10^{5}$

Table 3. Time parameters t substituted in equations 4–7 for the considered beam delivery and positron emission tomography imaging scenarios

	In-beam				Offline	
Accelerator	t_s (s)	t_p (s)	t_{irr} (s)	t_{end} (s)	t_{del} (s)	t_{meas} (s)
Cyclotron	60	0	60	40	300	1,800
Synch GSI	1	4	600	40	300	1,800
Synch HIT	1.5	1.5	180	60	300	1,800

Abbreviations: GSI = Gesellschaft für Schwerioneneforschung (Darmstadt, Germany); HIT = Heidelberg Ion-Beam Therapy Center (Heidelberg, Germany); Synch = Synchrotron.

signal by a factor of about 1.4 with respect to the 40-s time window that is currently exploited at GSI.

Spatial distributions of measurable decays. In terms of spatial distributions, the results presented in Figs. 7 to 9 for the most sensitive case of proton treatments show only a negligible dependence on the different irradiation modality within the same in-beam or offline imaging approach. On the contrary, a significant change of the activity map is observed between in-beam and offline imaging, regardless of whether biologic clearance is included. The former finding reflects the different spatial yield of activated ^{15}O and ^{11}C target fragments (Figs. 3 and 4), which correlates with the elemental distribution of oxygen and carbon in tissue. Besides the reduction of the measurable signal (note that variations of local concentrations may differ from the integral changes of Table 5), increasing biologic washout after isotope production further deteriorates the spatial correlation between activity and dose (Fig. 2). Especially in the case of head sites (Figs. 7 and 8), activation of well-perfused brain tissue is almost vanished in offline imaging. This results in a local activity maximum in the entrance channel because of high carbon concentration in subcutaneous adipose tissue, as observed in the work of Parodi *et al.* (12). Similar effects are found in paraspinal sites (Fig. 9), where offline images again basically reflect the ^{11}C activation map induced by proton treatments (Fig. 3).

For carbon ion irradiation, deformation of the activity map in dependence of the imaging strategy, elemental tissue composition and biologic clearance is expected to be less pronounced than in the proton case shown. In fact, at least (*i.e.,* for in-beam) 25% to 30% contribution to the overall measurable signal is attributed to the short- and long-lived ^{10}C and ^{11}C projectile fragments (Table 2, Figs. 5 and 6), which accumulate in the distal tumor region of interest for therapy monitoring.

DISCUSSION

Despite the simplified assumption of constant irradiation parameters and the approximate extrapolation to carbon ions, this study can offer a valuable insight on the available

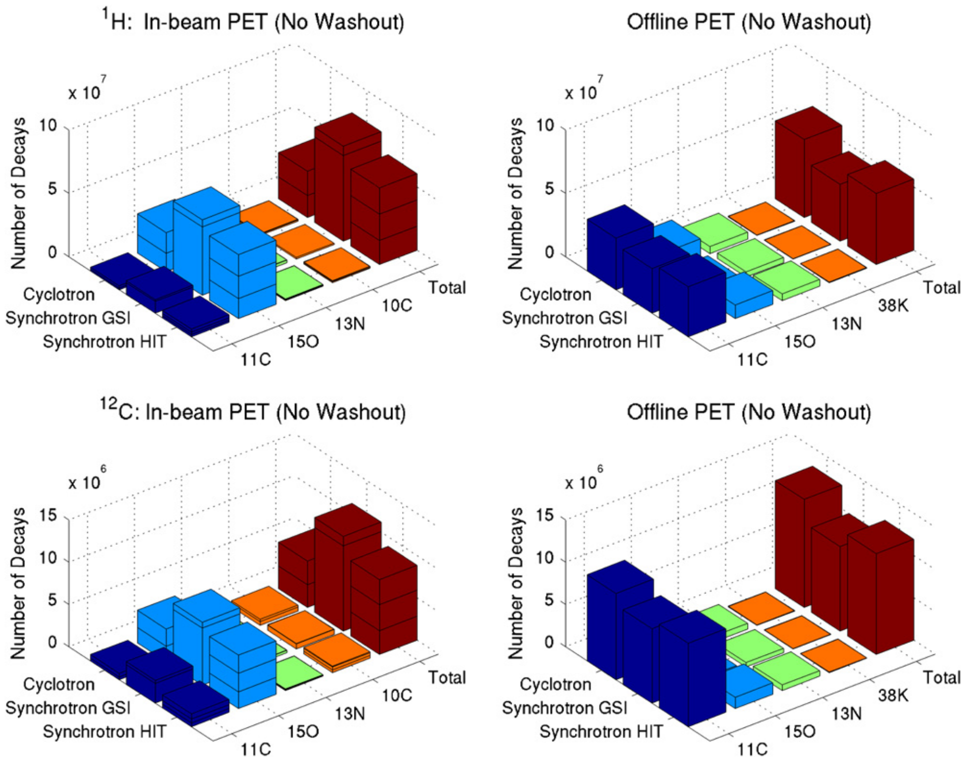

Fig. 5. Measurable physical decays from the main contributing ^{11}C, ^{15}O, ^{13}N, and ^{10}C (^{38}K) isotopes as well as total amount from all the calculated β^+-emitters for in-beam (left) and offline (right) positron emission tomography (PET) imaging of proton (top) and carbon ion (bottom) irradiation of a skull-base tumor (Fig. 2, left, and Table 2). Biologic considerations are not included. In the left panel, in-beam contributions decayed during beam extraction (cyclotron and Heidelberg Ion-Beam Therapy Centre [HIT] synchrotron), pauses (synchrotron only) and immediately after end of irradiation are separately shown from bottom to top (see the sub-division of the bars into "Lego bricks".). GSI = Gesellschaft für Schwerioneneforschung.

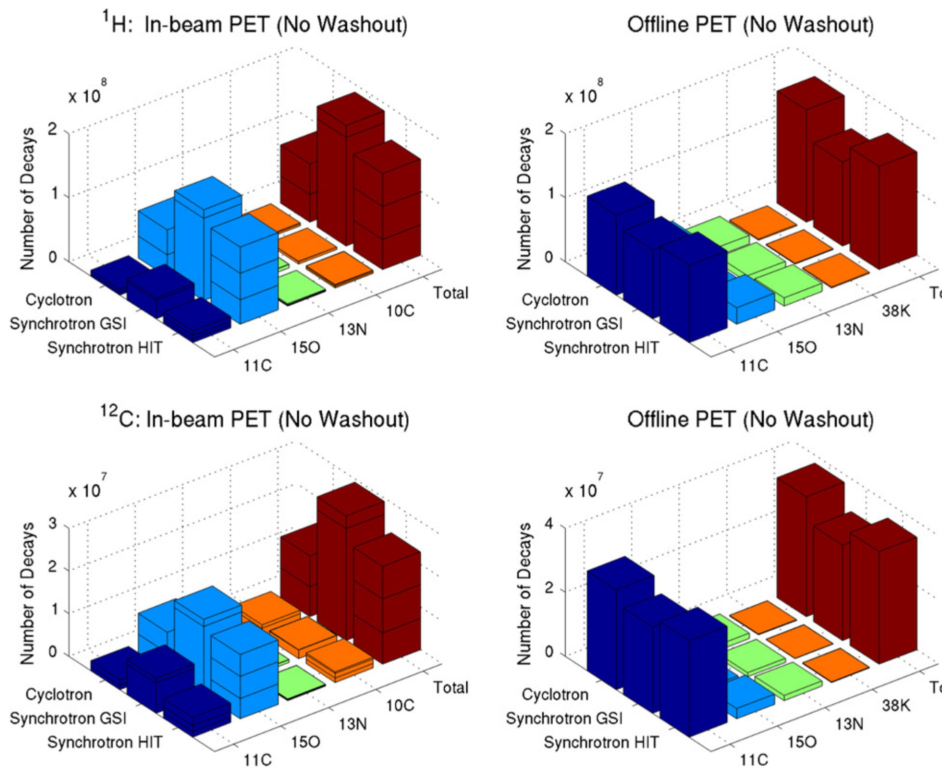

Fig. 6. Total amount with separate individual main contributions of the measurable physical (*i.e.,* without biologic considerations) isotope decays during in-beam (left) and offline (right) positron emission tomography (PET) imaging of proton (top) and carbon ion (bottom) treatment of a paraspinal tumor (Fig. 2, right, and Table 2). Again, different in-beam contributions are separately shown (Fig. 5). GSI = Gesellschaft für Schwerioneneforschung.

signal for in-beam and offline PET imaging at different ion beam facilities. In particular it shows that, for very short (<2 min) irradiation at conventional cyclotron-based facilities, suppression of random coincidences during beam delivery, and a few minutes of acquisition time immediately after irradiation are essential to compete with the amount of signal, which, *e.g.,* can be measurable offline for 30 min starting 5 min after treatment. For similarly short (~3 min) irradiation at the upcoming synchrotron-based HIT facility, in-beam PET imaging followed by 60-s acquisition after end of irradiation can provide a stronger or at least comparable signal as 10 times longer offline acquisition. This however requires efficient recovery of events decayed during the real ''beam-on'' time via random suppression techniques. The comparison

Table 4. Amounts of physical decays available for imaging at various facilities

			In-beam				
Ion	Site	Accelerator	Spill	Pause	End	Total*	Offline
^{1}H	Head	Cyclotron	$1.7\ 10^{7}$	—	$1.9\ 10^{7}$	$3.6\ 10^{7}$	$6.1\ 10^{7}$
^{1}H	Head	Synch GSI	$1.7\ 10^{7}$	$6.7\ 10^{7}$	$7.4\ 10^{6}$	$7.5\ 10^{7}$	$4.5\ 10^{7}$
^{1}H	Head	Synch HIT	$2.1\ 10^{7}$	$2.1\ 10^{7}$	$2.1\ 10^{7}$	$6.0\ 10^{7}$	$5.6\ 10^{7}$
^{12}C	Head	Cyclotron	$2.7\ 10^{6}$	—	$2.8\ 10^{6}$	$5.5\ 10^{6}$	$1.3\ 10^{7}$
^{12}C	Head	Synch GSI	$2.5\ 10^{6}$	$1.0\ 10^{7}$	$1.1\ 10^{6}$	$1.1\ 10^{7}$	$1.0\ 10^{7}$
^{12}C	Head	Synch HIT	$3.1\ 10^{6}$	$3.1\ 10^{6}$	$3.0\ 10^{6}$	$8.9\ 10^{6}$	$1.2\ 10^{7}$
^{1}H	Spine	Cyclotron	$4.3\ 10^{7}$	—	$4.8\ 10^{7}$	$9.1\ 10^{7}$	$1.8\ 10^{8}$
^{1}H	Spine	Synch GSI	$4.3\ 10^{7}$	$1.7\ 10^{8}$	$1.9\ 10^{7}$	$1.9\ 10^{8}$	$1.3\ 10^{8}$
^{1}H	Spine	Synch HIT	$5.2\ 10^{7}$	$5.2\ 10^{7}$	$5.1\ 10^{7}$	$1.5\ 10^{8}$	$1.6\ 10^{8}$
^{12}C	Spine	Cyclotron	$7.0\ 10^{6}$	—	$7.2\ 10^{6}$	$1.4\ 10^{7}$	$3.7\ 10^{7}$
^{12}C	Spine	Synch GSI	$6.6\ 10^{6}$	$2.6\ 10^{7}$	$3.0\ 10^{6}$	$2.9\ 10^{7}$	$3.0\ 10^{7}$
^{12}C	Spine	Synch HIT	$8.1\ 10^{6}$	$8.0\ 10^{6}$	$7.6\ 10^{6}$	$2.3\ 10^{7}$	$3.5\ 10^{7}$

Abbreviations: GSI = Gesellschaft für Schwerioneneforschung (Darmstadt, Germany); HIT = Heidelberg Ion-Beam Therapy Center (Heidelberg, Germany); Synch = Synchrotron.

In-beam events during the spills, pauses, and immediately after end of irradiation are separately reported.

*Total amount refers to the acquisition scheme (*i.e.*, in-spill decays are not measured at GSI, whereas a 90% and 100% recovery is assumed at the HIT synchrotron and at a continuous-wave cyclotron, respectively).

Table 5. Measurable decays for in-beam and offline positron emission tomography imaging of proton irradiation when taking biologic considerations into account (no washout vs. washout)

			In-beam		Offline	
Ion	Site	Accelerator	No washout	Washout	No washout	Washout
^{1}H	Head	Cyclotron	3.6 10^{7}	2.4 10^{7}	6.1 10^{7}	2.7 10^{7}
^{1}H	Head	Synch GSI	7.5 10^{7}	4.4 10^{7}	4.5 10^{7}	1.9 10^{7}
^{1}H	Head	Synch HIT	6.0 10^{7}	3.8 10^{7}	5.6 10^{7}	2.4 10^{7}
^{1}H	Spine	Cyclotron	9.1 10^{7}	7.2 10^{7}	1.8 10^{8}	1.0 10^{8}
^{1}H	Spine	Synch GSI	1.9 10^{8}	1.3 10^{8}	1.3 10^{8}	7.2 10^{7}
^{1}H	Spine	Synch HIT	1.5 10^{8}	1.1 10^{8}	1.6 10^{8}	9.2 10^{7}

Abbreviations: GSI = Gesellschaft für Schwerioneneforschung (Darmstadt, Germany); HIT = Heidelberg Ion-Beam Therapy Center (Heidelberg, Germany); Synch = Synchrotron.

becomes even more in favor of in-beam PET for synchrotron-based deliveries with longer irradiation times and extensive beam pauses, as is currently the case at GSI. In all scenarios, in-beam PET imaging enables detection of the significant activity contribution from the ^{15}O yield (Figs. 5, 6), which, in the proton cases considered, was found to equal the ^{11}C production and to have a more homogeneous distribution in tissue (Figs. 3, 4).

Furthermore, in-beam imaging is less affected by the degradation of the signal resulting from washout (Table 5, Figs. 7–9).

For clarity, the data analysis was limited to single-field irradiation at a typical physical dose of 1 Gy. The results can be scaled approximately linearly with the number of fields and applied dose, provided that the duration of each irradiation and the time interval between the individual portals are short in comparison to the isotope half-lives and similar for the different facilities. For example, doubling the dose with a 2-min interval between redelivery of the same fields would increase the offline PET signal (physical decay) by a factor of 1.6 to 1.8. Of course, in-beam PET imaging of multi-field irradiation can provide separate detection of the first field as well as cumulative distribution of all the portals. The latter can be obtained from co-registration and superimposition of the individual acquisitions taken in the different treatment positions. On the contrary, the offline approach can only image the integral activation from the whole treatment. Depending on the overlap of the fields, this may result in loss of information on the distal range. Furthermore it can imply a severe deterioration of the signal formed by the first delivered fields, depending on the time elapsed between irradiation and imaging (12).

The data shown represent the number of decays that are potentially available for detection. Assuming an attenuation of 80% and a detection efficiency of 4%, a total of 10^{7} to 10^{8} decays would correspond to 80,000 to 800,000 coincidences to be collected over the whole acquisition, consistently with the numbers quoted in Crespo *et al.* (18) and observed in Parodi *et al.* (12). The achievable counting statistics is therefore far below the typical values of nuclear

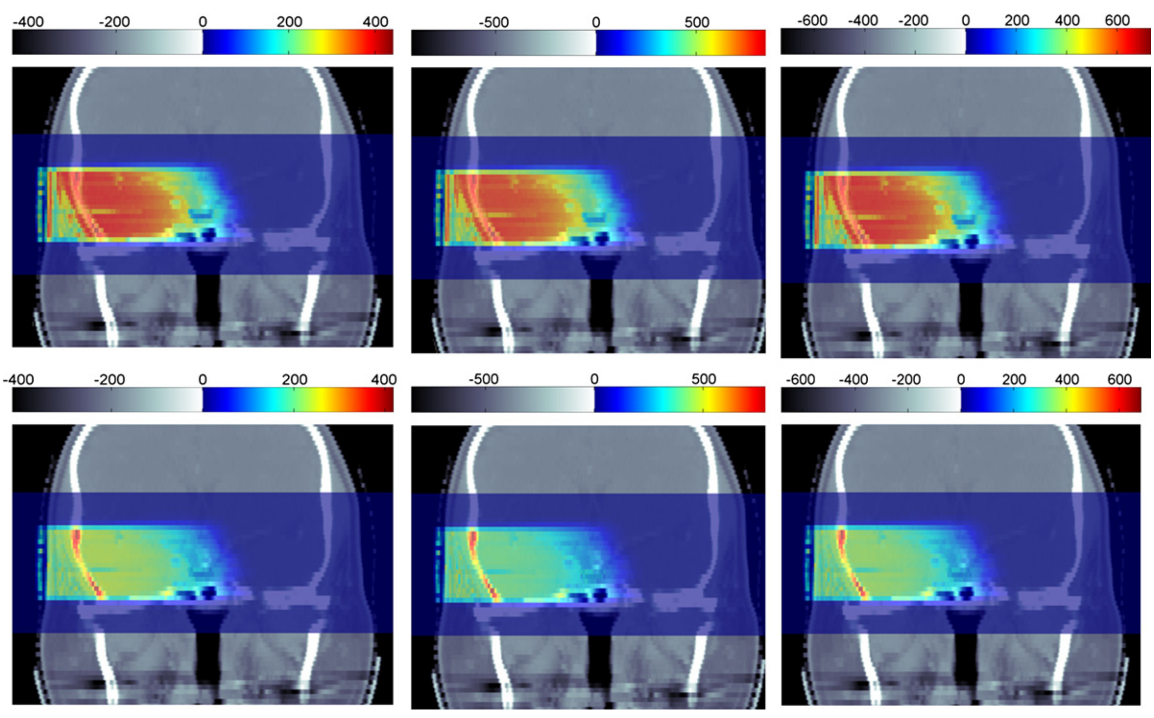

Fig. 7. Spatial distribution of isotope decays available for in-beam positron emission tomography (PET) acquisitions during proton irradiation of a cranial-base tumor (Fig. 2, left) at cyclotron- (left) and synchrotron-based (middle, Gesellschaft für Schwerioneneforschung [GSI]; right, Heidelberg Ion-Beam Therapy Centre [HIT]) facilities. Calculations exclude (top row) or take into account (bottom) biologic considerations. Color bar indicates decays concentrations in units of 10^3/ml. Grey scale depicts arbitrarily rescaled CT numbers.

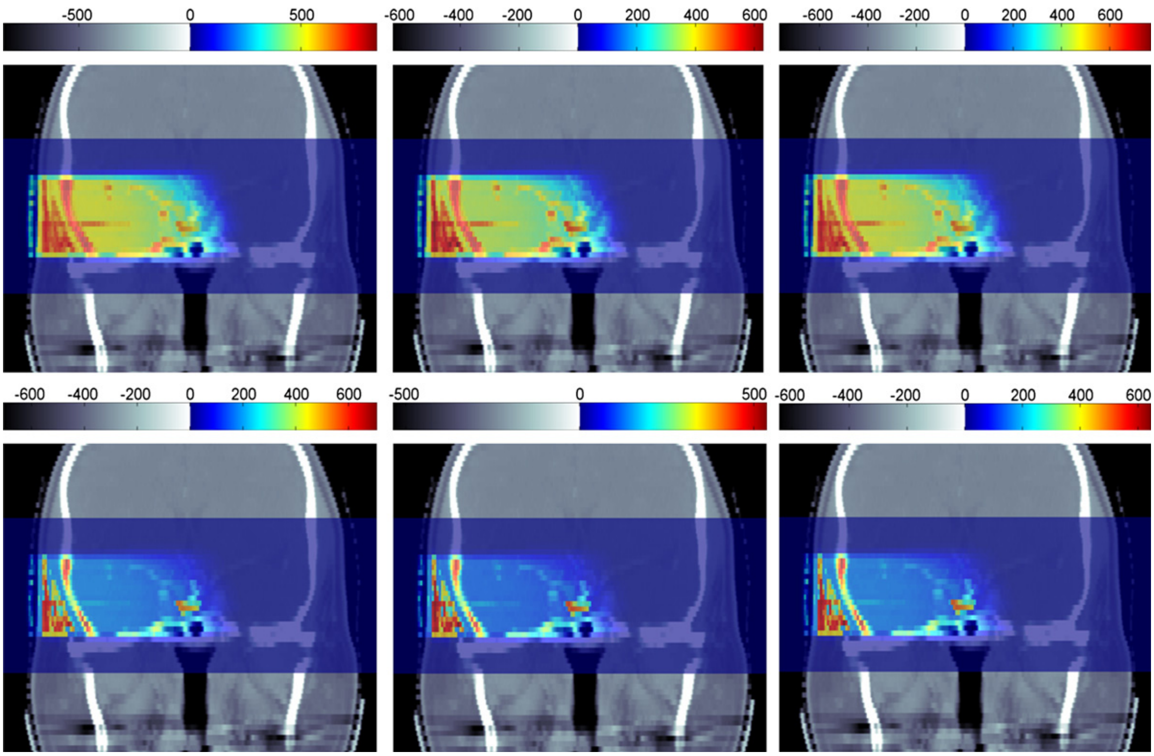

Fig. 8. Calculated spatial distribution of isotope decays to be detected offline after proton irradiation of a cranial base tumor (Fig. 2, left) at cyclotron-based (left) and synchrotron-based (middle, Gesellschaft für Schwerioneneforschung [GSI]; right, Heidelberg Ion-Beam Therapy Centre [HIT]) facilities when excluding (top row) or taking into account (bottom) biologic considerations. Interpretation of grey scale and color bar is the same as for Fig. 7.

medicine tracer imaging. Thus even a few percent reduction of the available signal in dependence of the acquisition strategy may have significant consequences on the quality of the reconstructed images and related usable information for the purpose of treatment verification.

Although attenuation and scattering of the annihilation photons in the patient are the same for all imaging strategies, detection efficiency and imaging performances are strongly dependent on the choice of the detector system. Because of the typically small (~16 cm) axial coverage of commercial scanners, large-area dedicated detectors may still be competitive in terms of detection efficiency. Instead, limited-angle artifacts irreversibly compromise the achievable image quality almost independently from the available statistics, thus calling for careful design. Both aspects were exhaustively covered in the report by Crespo *et al.* (18) and therefore omitted in this study. Finally, for true in-beam acquisitions during the ''beam-on'' time, an ideal 100% recovery of signal was considered as the ''best case'' scenario at continuous-wave cyclotrons. A more realistic value (*e.g.,* 90%) for the MGH cyclotron would lower the discussed ratios between in-beam and offline measurable decays by 3% to 5%. For the HIT synchrotron, the assumption of 90% in-beam signal recovery was based on previously reported experimental data (16, 17). In both scenarios, in-beam random suppression techniques will also enable detection of millisecond-short–lived emitters (*e.g.,* ^{8}B, ^{12}N), which have typically higher β^{+}-endpoint and corresponding longer positron range in tissue. However, similar to the report by Crespo *et al.* (17), identification of this contribution and evaluation of its impact has been omitted from this work because of lack of experimental information.

CONCLUSIONS

We have presented a comparison of the measurable in-beam and offline PET signal for single treatment fields of head-and-neck and paraspinal tumor sites, considering different beam delivery and imaging approaches at cyclotron- and synchrotron-based ion beam facilities. These data can assist other institutions in evaluating the optimal implementation of PET quality assurance in ion therapy. The values must be interpreted taking into account the specific fractionation scheme and the properties of the PET imaging systems under investigation, with particular emphasis on imaging performances and detection efficiency.

Under the assumption of the same dose delivery and comparable detector equipments, this study indicates a 15% to 20% loss of counting statistics for in-beam imaging at an efficient synchrotron-based facility such as HIT in comparison to the current situation at GSI. Nonetheless in-beam PET acquisition at HIT is still expected to provide a stronger, or at least comparable, signal with better correlation with the delivered dose (especially for light ions like protons) in comparison to offline approaches. This estimation however postulates prolongation of the in-beam acquisition

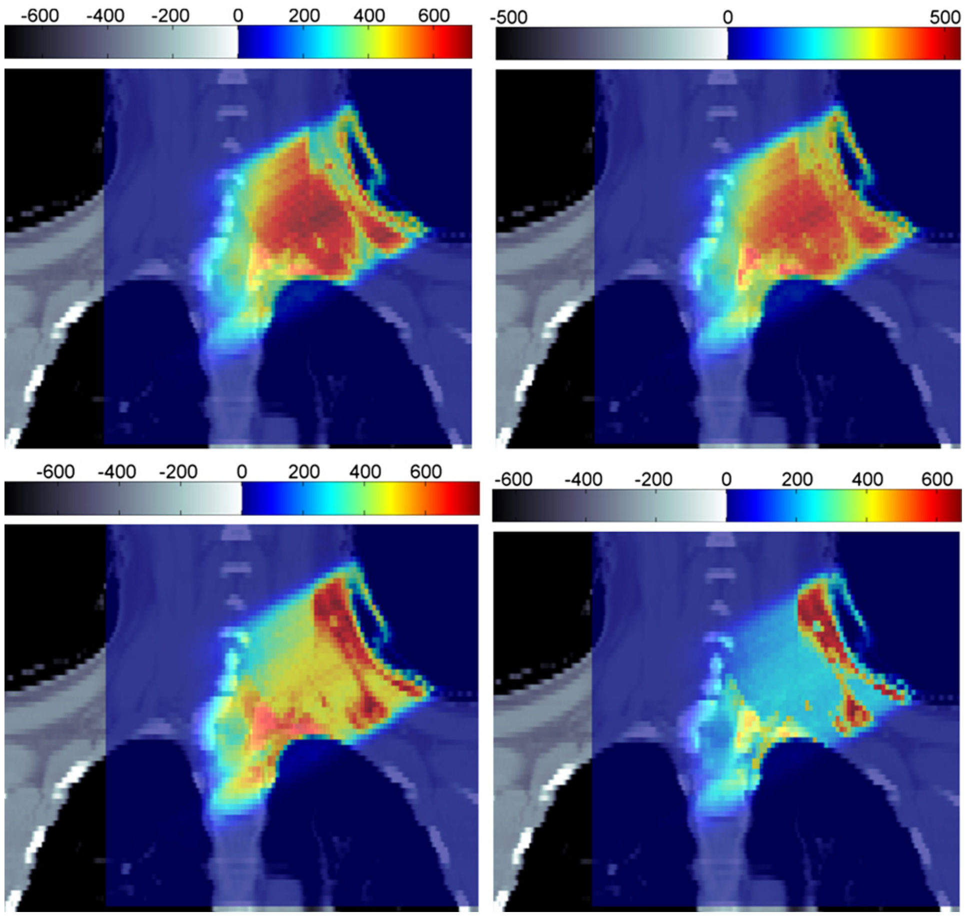

Fig. 9. Calculated spatial distributions of isotope decays to be detected in-beam (top) and offline (bottom) for proton irradiation of a paraspinal tumor (Fig. 2, right) at the Heidelberg Ion-Beam Therapy Centre [HIT] synchrotron, excluding (left) or taking into account (right) biologic considerations. Color-wash display is in units of 10^3/ml. Grey scale depicts arbitrarily rescaled computed tomography numbers.

by ~1 min after the end of irradiation as well as implementation of an efficient method for recovery of the decays occurred in the pulses of the beam delivery. In this latter respect, experimental investigations addressing the role of millisecond-short–lived emitters and the impact of their longer positron range on the imaging performances will be required.

For very short irradiation times such as those typically achieved at conventional cyclotron-based facilities with passive beam shaping systems, a few minutes of acquisition time directly after end of irradiation will be necessary for counting statistics. Thus the expected benefit of in-beam PET imaging has to be carefully analyzed against the penalization of efficient patient throughput.

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